Portfolio: Nano-Enhanced Battery Thermal Management System

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Problem Statement

Electric vehicle batteries generate significant heat during operation, which reduces performance and lifespan. This project focuses on designing a cooling system using an Oscillating Heat Pipe (OHP) combined with liquid and nano particles (Al_2O_3).

Research Overview

This research focuses on solving the overheating issues in high-capacity lithium-ion batteries. The system uses L-shaped oscillating heat pipes (OHP) together with Al_2O_3 nanofluids to improve heat dissipation, enhance cooling performance, and help prevent thermal runaway.

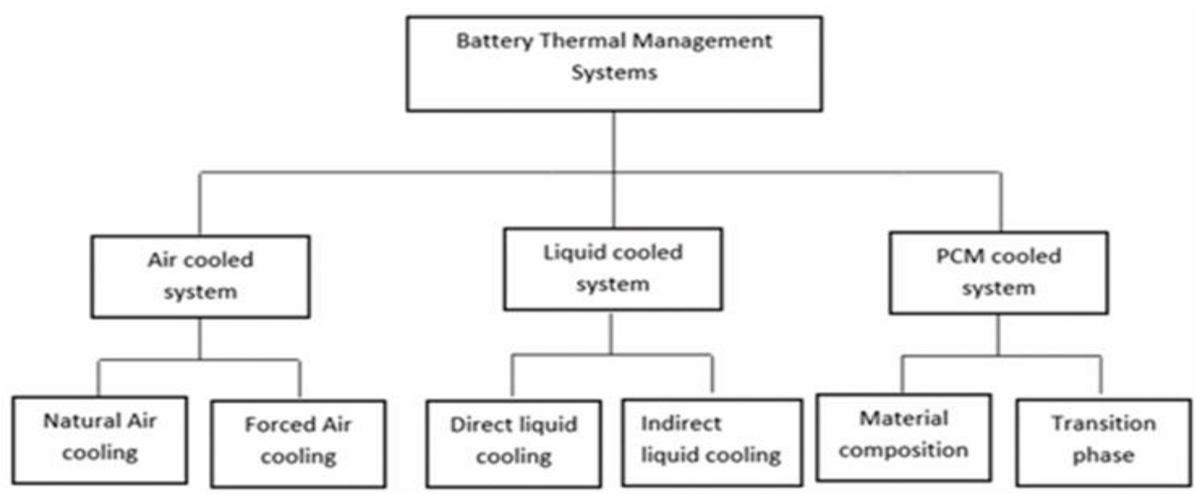

Figure 1: Classification of common Battery Thermal Management Systems.

Design Approach (SolidWorks)

Designed L-shaped oscillating heat pipe geometry

- Heat is collected from the bottom of batteries by heat collector plates, distributed along the heat pipe, and directed to the inside of the liquid
- Cooling plates is used not to touch batteries and the oscillating heat pipe
- Developed a full assembly model for thermal system simulation

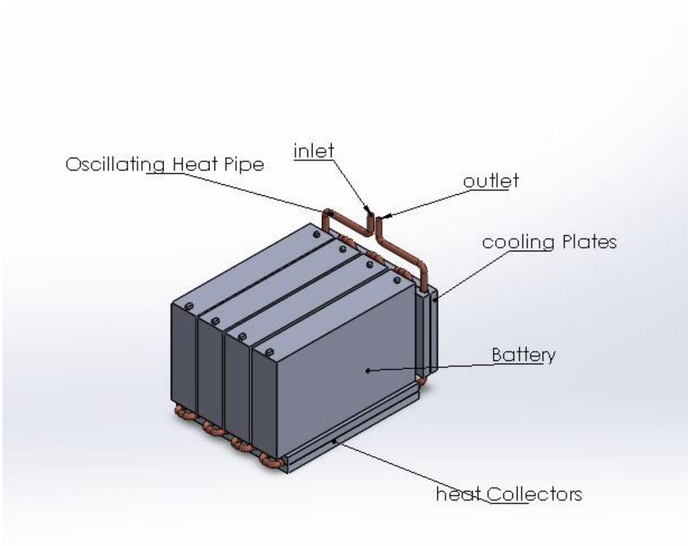

Figure 2: SolidWorks assembly showing the OHP integration with battery modules.

Methodology

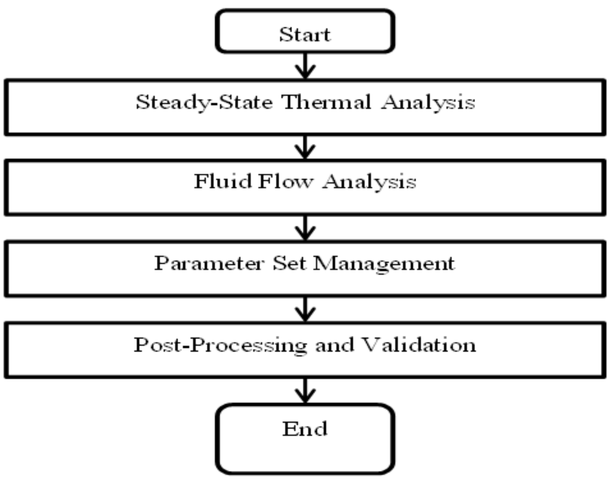

Figure 3; Research Flow Chart

The methodology links the thermal and fluid domains. It is essential to show that these two domains affect each other in thermal management systems. This can be done using the System Coupling module in ANSYS Workbench. This coupling allows ANSYS Mechanical and Fluent to transfer data.

Coupling Steady-State Thermal and Fluid Flow In ANSYS

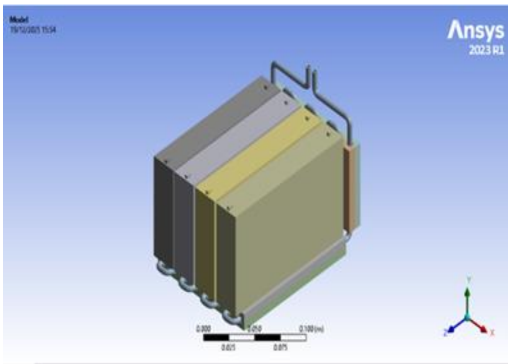

Figure 4; Geometry Modelling

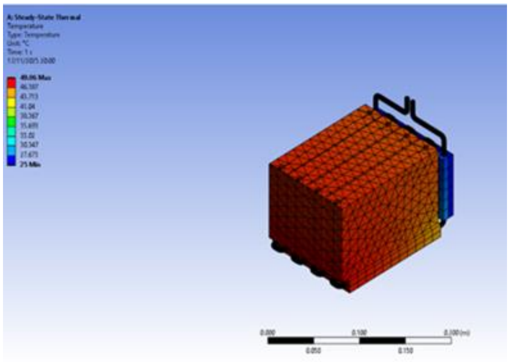

Figure 5; Thermal Modelling

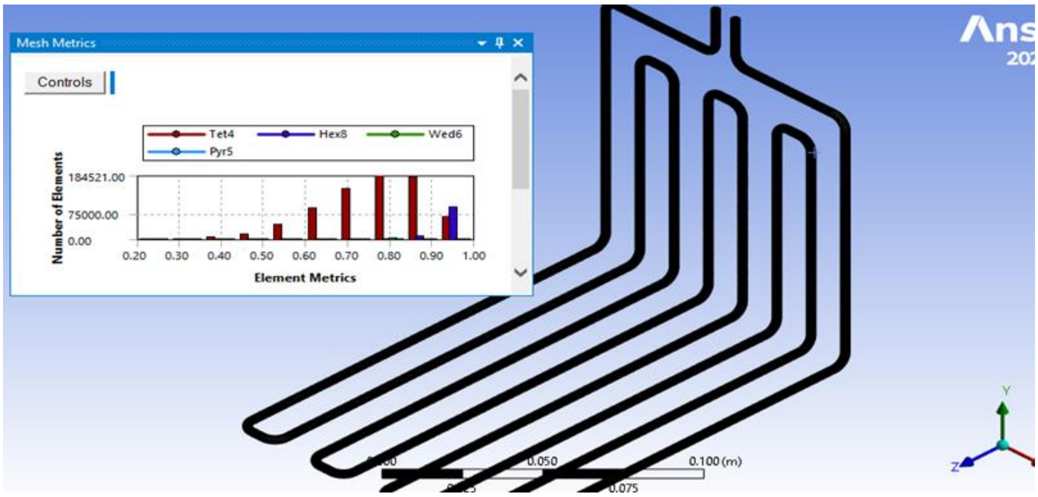

Figure 6; Meshing for Oscillating Heat Pipe

The geometry is imported from SolidWorks into the Steady State region of ANSYS, as shown in Figure 4.

1. Define the materials for the oscillating heat pipe (copper), Heat collector and cooling plate (aluminium), and battery (lithium-ion).
2. Split the fluid and pipe, and meshing (automatic meshing) all solid bodies.
3. Calculate all the heat from batteries and apply it in steady-state conditions.
4. Use all results of heat flow to the heat pipe from steady-state conditions as the boundary conditions of fluid flow.
5. Meshing for the oscillating heat pipe shown in Figure 6 is important because, in this region, consider only the fluid wall.

Governing Equation

Flow is Laminar flow is laminar because the pipe diameter is 3mm. Before Meshing, use the continuity equation for mass, the momentum equations for pressure and density of fluid, and the energy equation for fluid temperature. After meshing, apply the discretization equation for each cell of the mesh.

Numerical Simulation (ANSYS Fluent)

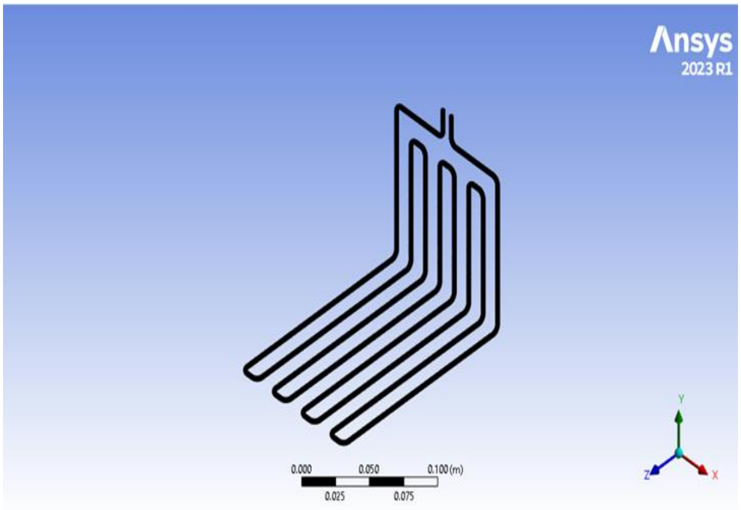

Figure 3: Discretized domain showing the conformal mesh at the solid-fluid interface.

Computational Fluid Dynamics (CFD) analysis was carried out to study the heat transfer behaviour of the system. The computational mesh was carefully optimized, achieving an orthogonal quality of 0.797, which ensures good accuracy and reliable simulation results.

Results and Thermal Performance

The inlet temperatures of the water and nanofluid are 283K,293K, and 298K, the outlet temperature also reaches a maximum. The flow velocity is 0.1, 0.2, 0.3, and 0.4 m/s. Simulation results verified that although the highest exchange temperature is 6K for water, the sudden temperature rise, as shown in figures 1, 2,3 used water. For nano particle involve fluid, the maximum exchange temperature is nearly 4K and the minimum 1K. There is no sudden increase and decrease for nanofluid. At moderate velocities of 0.3, as shown in figures 4, 5, 6, nanofluids were used to get a stable temperature.

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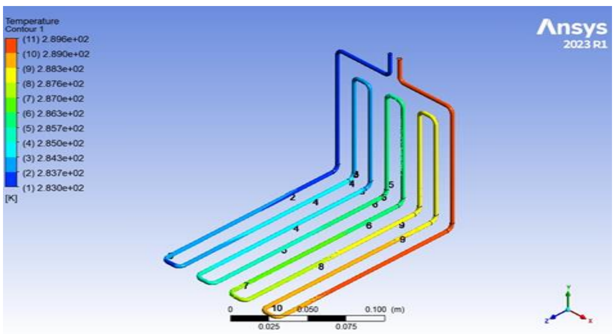

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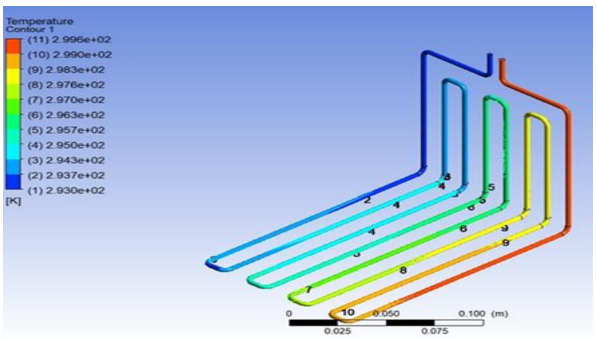

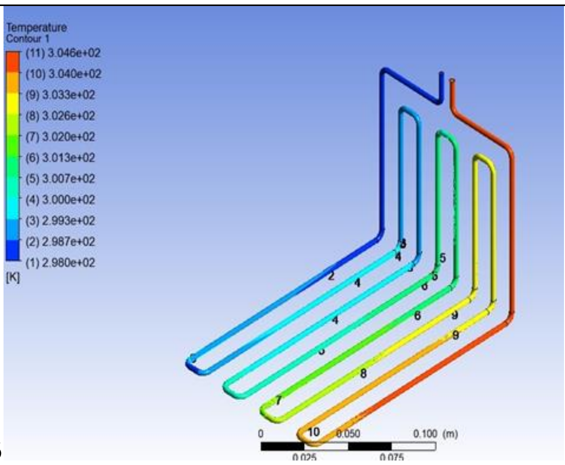

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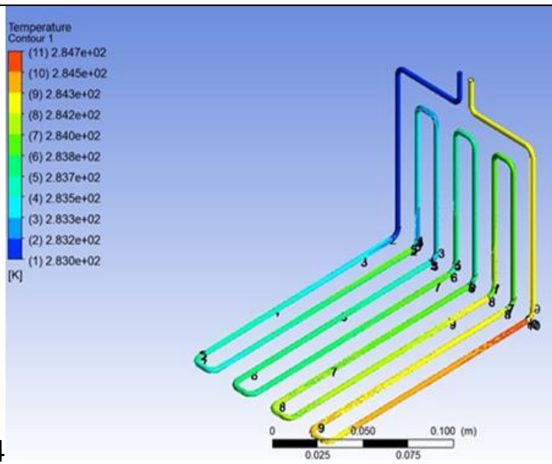

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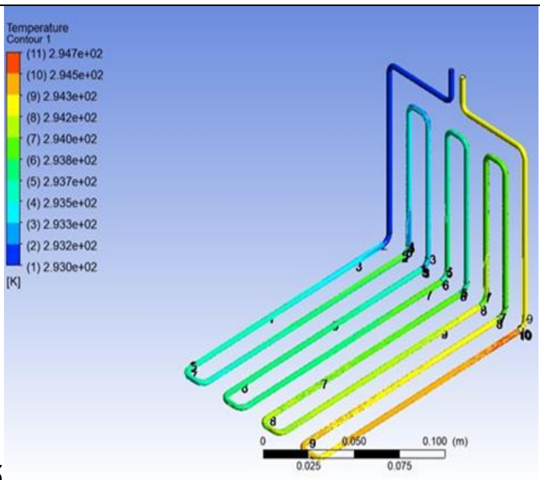

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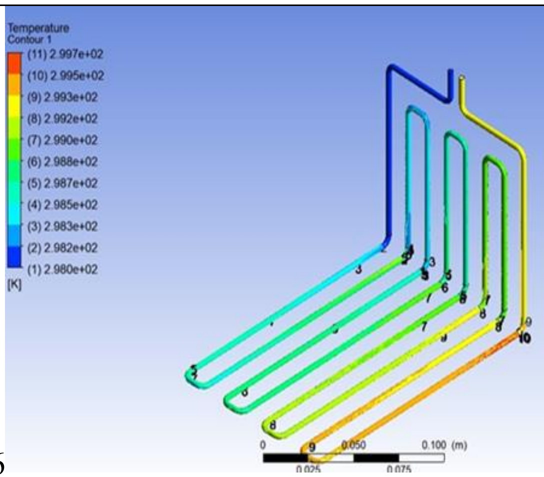

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Conclusion

The higher nanoparticle concentration increases both thermal conductivity and heat capacity. At moderate velocities of 0.3 and 0.35m/s, the nanofluid exhibits more stable performance compared to water. When comparing 0.5 % and 0.25% nanofluid, the higher thermal conductivity improves heat exchange from the oscillating pipe wall and becomes stable heat transfer.